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Partially Indexed Image Corpora Access Through Textual And Visual Retrieval Process

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Abstract Large collections of images are seldom accurately indexed with keywords. Pure visual retrieval, however, may not give good results when the corpus is domain-independent. This paper addresses the issue of efficient retrieval from image corpora in which only a little proportion is textually indexed. Our hybrid approach integrates textual querying/search with content-based retrieval.

A preliminary double classification of the corpus is achieved and an adequate retrieval process is designed to take advantage of user-system interaction, via relevance feedback mechanism. Experimental results with the PICAP prototype are reported and discussed to demonstrate the effectiveness of this approach.

1 Introduction

Whereas collecting thousands of images is now possible, an accurate and comprehensive image indexing is still a time-consuming and an error-prone process. We claim, however, that *partially indexed* corpora can be organised in a way that allows interesting retrieval performance.

Two main categories of image retrieval systems exist. Systems allowing only *keyword querying* only take into account manually indexed images, and performance will depend on the quality of indexing. Many indexing schemes are available, from classical keyword vectors [10,25], to more complex data structures [9,7,20]. Of course, the difficulty of the indexing task varies accordingly. *Content-based retrieval* systems use visual similarity to retrieve relevant images [26]. The main advantage of this approach is that features can be computed off-line [30,12,27,15].

We notice that content-based systems give little attention to the semantic part of the query. However, we believe that keywords remain an important mediating object between users' desires and image content. Therefore, we propose a new retrieval strategy based on a prior corpus organisation that takes into account both textual and visual aspects. Our work applies to domain-independent corpora, in which few images are actually manually indexed. This is crucial, as image corpora easily include large amounts of images.

This paper includes a brief description of some related works (§ 2). We examine in § 3 how a corpus may be characterised and organised. Then we

show how a retrieval process can be designed to take advantage of this organisation (§ 4). Finally, current experimentations and results are presented and discussed (§ 5).

2 Related work

Drawbacks of pure visual retrieval systems have already been pointed out [1] and the combination of visual and textual features has been addressed in several ways. Those hybrid approaches try to fulfill a double objective of textual and visual relevance.

Working on a restrictive domain helps achieving good results. For example, locating and naming faces on images or movies is possible by combining very specialized visual features with textual features extracted from captions [8,23]. When no particular assumption can be made on the domain, textual description can be attached to images by means of keywords, and pictorial content can be characterized by visual features, whereas no link is established between these two different spaces: some systems rely on keywords, shape, colour and texture to build a similarity measure [16]. Textual annotations have to be provided by hand. As an attempt to bring together textual and visual properties, keywords can be associated to a combination of visual features [17]. These associations are not established automatically. In the web context, images come with texts in HTML pages. Assuming that the text is related to the content of images, the latent semantic indexing technique allows the automatic computation of descriptors including both

textual and visual features [24]. Other statistical methods can be utilized in order to associate words to images, based on some learning processes [14].

These approaches do not consider partially indexed corpora. They give little attention to feedback mechanism. We believe, however, that valuable information can be extracted from user interaction.

3 Corpus organisation

In our approach, images are characterized on a visual and *possibly* textual point of view. Indeed, not all images need to come with a textual annotation, whereas some visual features are automatically computed on each image. Our first goal is to select some relevant features, that can highlight image similarities, and be used thereafter for classification purposes.

3.1 Textual description of images

We assume that the collection of images is partially and roughly indexed with text. In other words, a small number of images are associated with textual description, and, when this description is available, three or four words only are sufficient.

The lexical reference system WordNet [6] is used as a knowledge source to help dealing with synonyms, namely. In WordNet, words are organised into synonym sets called "synsets". Each synset represents a lexical concept and is connected to other synsets with different kinds of semantic relationships that provides access, for example, to "generic", "specific", "antonym" concepts.

The classical vector model and its associated cosine measure have been chosen as a textual similarity measure [22], and will be used for both clustering and retrieval.

3.2 Visual description of images

Image analysis techniques provide us with many possible features to describe colour, texture or shape [19]. Each image can be automatically characterized by a set of features, but numerous problems remain. For example, it is well known that some features are particularly well adapted to a given domain, whereas they achieve poor results in other ones. Useful visual features are close to human perception, in order to take easily into account user judgements [18]. In our environment, we cannot assume domain-dependent knowledge that could help selecting relevant features, or defining specific ones, and thus we have to use "generic" features.

As a very general — though powerful — feature, we chose colour histogram [28], represented in L*u*v* colour space. L*u*v* is a device-inde–pendent and perceptually uniform colour model, thus well suited for comparisons. We considered three different colour spaces: RGB, L*u*v*, HVC. Not surprisingly, L*u*v* achieved the best results for colour comparisons.

A subset of 128 representative colours can be selected from the entire corpus, so that each image can be quantized according to this reduced colour map, and represented by a 128-components vector.

To improve localisation and to allow "layout" comparison, features are not only computed over the entire image, but also on small image areas: a fixed grid is applied on the image, defining 32x32 pixels squares called *tiles*. A colour histogram is computed for each tile of each image in the corpus. The spatial organisation of visually homogeneous areas (according to the colour feature) provides for a rough sketch of the "shapes" contained in the image.

This basic visual characterization is intented to test the validity of our approach. As a second step, we are planning to introduce more powerful features, such as texture and shape, that will capture other visual properties of images. These are, however, difficult issues. On one hand, extraction of the "relevant" shape is not straightforward [1]. Many approaches exist, some of them being more adapted to object recognition purposes (*i.e.*, when "known objects" have to be carefully characterised, before attempting to identify them in various images). On the other hand, while it is easy to give a rough description of texture, we believe that fine texture characterization has to be achieved on homogeneous zones. In both cases, some kind of image segmentation will be required.

3.3 Classification-based corpus organisation

For corpus organisation purposes we focused on clustering techniques [4], which aim at grouping similar objects. These techniques have been extensively used and studied in the field of Information Retrieval [11, 29, 5, 31]. It

is also worth noting that these techniques have been mainly applied to textual information organisation; we intend to use them for *image* classification as well, based on visual features. In our approach, agglomerative hierarchical clustering (AHC) has been chosen to classify the corpus. Other works suggested alternative classifications of images based on visual features [13].

Two classifications have been constructed over the corpus: a textual and a visual classification, based on features described in sections 3.1 and 3.2 above; the resulting structure is called "dendrogram".

The textual dendrogram includes only textually indexed images, that is, potentially few images. The visual dendrogram, however, includes all images, as the colour feature computation can be carried out automatically.

The advantage of hierarchical structure is that the cluster size can be controlled by a cut-off value (cf. fig. 1): at lower levels, clusters contain few, very similar images. The cluster size grows as we consider higher levels, and this suggests a simple way to balance precision against recall (*Precision* is defined as the proportion of the retrieved documents which are relevant, whereas *recall* is the proportion of relevant documents that are actually retrieved). The cut-off value is determined during retrieval session, and is thus context-adaptive.

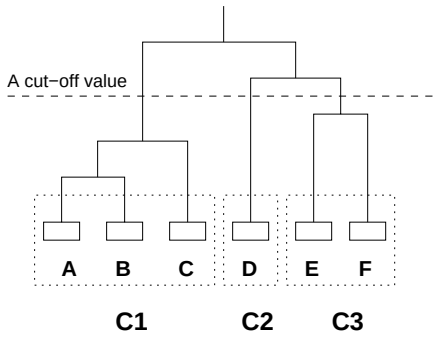


Fig. 1 The dendrogram: a hierarchic structure produced by Agglomerative Hierarchic Clustering.

4 Image Retrieval Process

4.1 Process outline

In our point of view, textual query the best way for the user to launch a retrieval process. Of course, the visual aspect is important for image search, but it seems difficult to assume that the user has already found a good example of the image(s) he/she is looking for or that he/she is able to sketch the desired images.

Figure 2 shows the overall image retrieval process. The initial textual query allows the retrieval system to propose a first set of images. An important characteristic of our approach is that this initial image set contains both textually indexed and not indexed images thanks to the tunnel mechanism (§ 4.3).

The user can then express his visual needs or preferences by providing precise feedback. This is performed by giving examples and counter-examples of what is relevant in respect with textual aspect and visual as-

pect (colours, layout...). Indeed, users find it more comfortable to provide interactively examples of relevant and irrelevant images. This feedback is exploited by the system in order to build a *virtual image* which gathers its understanding of the user's current need.

A graphical interface allows user to pick words and to browse through synsets, using *generic/specific*, *synonym* and *part of* relationships to navigate. The vocabulary is limited to nouns from *WordNet*. Each selected item is then qualified : "absolutely", "rather" or "possibly" *present* or *absent* in the image.

Section 4.3 describes the cluster collecting process and our image ranking scheme. Section 4.2 explains how the virtual image constitutes a support for the relevance feedback mechanism (feedback criteria and modalities will be given). To improve the legibility of the paper, we present this section before the collecting and ranking section (since the latter operation, in the general case, assumes a visualisation phase and user's feedback gathering).

4.2 Virtual image as a support for relevance feedback

After a visualisation step, it is important to gather information as precise as possible from user. Actually, images may have been judged relevant in only one point of view (textual or visual): the system needs to know in what extent a retrieved image is relevant or not to the user. To achieve this, we propose a twofold judgement on each image.

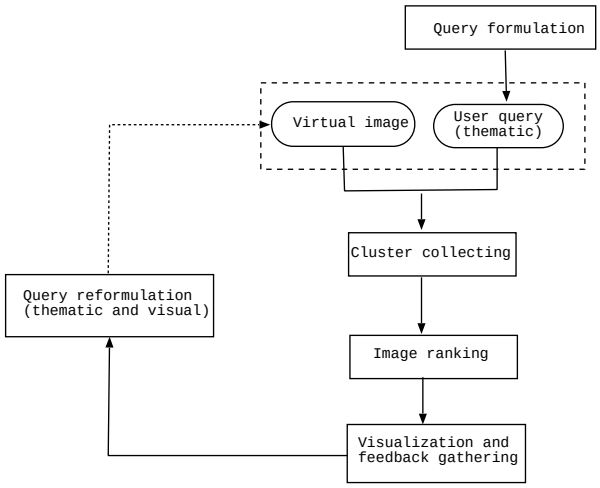


Fig. 2 The overall image retrieval process

The theme. – This kind of judgement is used to determine what themes are to be searched or avoided. A relevant image may feature new interesting themes that should be included in the next query, by means of reformulation.

The colour. – Colour has been chosen as the most representative among visual indices. It is a fundamental visual property of images, as users are very sensitive to colour; it has also a great discriminant power [28].

For each judgement type, user can "accept", "reject" or have "no idea" about each selected image, and can moderate his judgement with a weight ranging from 1 to 10.

Furthermore, **the layout** seems to be an additional good criterion candidate, as it captures the whole "composition" of the image, and then some shape properties. In this work, "layout" refers to spatial localisation of blobs, each blob featuring some homogeneous visual feature. The user has the possibility to choose, among the proposed images, the most representative one

(if any) according to the layout. This "typical" layout will be used for image ranking purposes at the next retrieval step.

From this judgement data, the system derives a representation of its understanding of user's needs. Practically, we are looking for discriminating information : we will try to find both colours and themes that are in common in the set of relevant images, and then in the set of irrelevant ones.

To handle this information, we introduce the **virtual image** concept. From the user feedback and from the features associated with displayed images, we can build two parts for wanted and unwanted image features, respectively. This applies both on textual and visual features. The reformulation allows the "filling" of these different parts of the virtual image as follows:

- **textual reformulation**: the thematic positive (resp. negative) feedback can be handled by a simple weighted averaging of the vectors representing the index of the user-relevant (resp. irrelevant) images.

- **visual reformulation**: as image colours are also represented by vectors, the same method can be applied to handle the positive and the negative feedback on the colour aspect as for textual reformulation.

4.3 Cluster collecting and image ranking

The retrieval process consists in searching the textual dendrogram and the visual one in order to select some clusters likely to contain relevant images. The selected images are then ranked before being presented to the user.

Searching the textual or visual dendrogram is done by an ascendant method which performs a constrained generalisation of the low level clusters (adapted from [21]). We use constraints that control the minimum/maximum size of the cluster, its dispersion rate, and the maximum distance. Depending on the collecting result, these constraints can be weakened.

Step 1 – Textual dendrogram search. — The thematic hierarchy is searched using the textual part of the virtual image (cosine measure is evaluated between "positive" part of the query and the cluster centroid). Depending on clusters size, though, not all images may be equally relevant to the query: this motivates an additional textual ranking process that will be used for the *tunnel* mechanism (see below). Let us suppose that cluster $C1$ is selected, in which image A is the best ranked.

Step 2 – Visual dendrogram search (direct). — The visual hierarchy is searched using the visual (colours) part of the virtual image (The "positive" part of the query and the cluster centroid are matched). This results in some likely interesting clusters.

Step 3 – Visual dendrogram search using tunnels. — The corpus organisation process (§ 3) has led to two different images classifications, and we believe that links can be established between the thematic classification and the visual one, yielding to a textual and visual synergy, which is based on the following hypothesis: "Textual similarity and visual similarity are not independent". Figure 3 illustrates this textual and visual cooperation called *tunnel*.

We assumed that image A from thematic cluster $C1$ is particularly relevant (step 1, above). The tunnel hypothesis tells us to consider visually similar images, which consists in locating image A in the visual dendrogram, and to consider that some "neighbourhood" around A as relevant. In our example, A also belongs to visual cluster $C2$, and all images in $C2$ will be selected as likely relevant (visually and textually). Some of them are indexed whereas others are not.

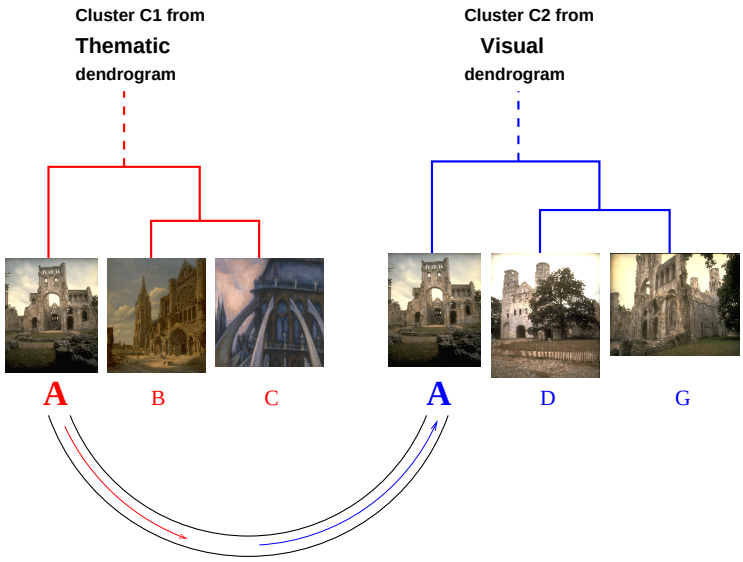


Fig. 3 The "tunnel" mechanism.

The overall advantage of this collecting process is that **non-indexed images have a chance to be retrieved** thanks to the tunnel mechanism even at the first retrieval attempt (*i.e.*, with a textual query). Retrieved images in all the selected clusters can now be **ranked** in a unique list. The following formula is used to rank images:

$$S = w * S_t + (1 - w) * S_v.$$

where the visual score (S_v) and thematic score (S_t) may be weighted by the user (w factor) so that it is possible to give more or less importance to visual search against textual search.

This ranking uses both positive and negative parts of the virtual image *i.e.,* what is wanted or unwanted by the user. Again, the cosine measure can be used for S_t and S_v after merging the positive and negative parts of the query in one vector (by affecting positive weights to wanted criteria and negative weights to unwanted ones).

Furthermore, if a typical layout has been chosen by the user, retrieval precision can be considerably improved by introducing a local visual matching (as opposed to the global colour-based matching). The term S_v above is then defined as a linear combination between a local term and a global term (the w_l factor gives less or more importance to the local visual term against the global one):

$$S_v = w_l * S_{l,v} + (1 - w_l) * S_{g,v}.$$

5 Experiments

5.1 Corpus characteristics and indexing

Experiments reported in this paper are based on an image corpus containing 2470 heterogeneous images (pictures, paintings, ...). In fact, almost every image is indexed with text, but the conducted experiments aim at evaluating

the impact of partial indexing on retrieval in our approach. Therefore, we added to the PICAP prototype the facility to "hide" some of the textual indexing.

In this context, the text descriptions to hide have been chosen randomly, or by hand. *Random indexing* consists in choosing randomly a certain amount of images, and to make their indexing keywords "visible" to the system. *Ad hoc indexing* relies on manual selection of images for which indexing should be kept visible. We define three different indexing conditions:

- *Random indexing* of 20% of the total amount of images.
- *Random indexing* of 40% of the total amount of images.
- *Ad hoc indexing* of 5% of the total amount of images.

It is worth noting that *ad hoc* indexing concerns far less images than random indexing: this is on purpose, since we want to prove that our approach can achieve good performance, provided few images — chosen by hand — are indexed.

5.2 Experiment description

To assess performance of the textual and visual approach, we let the user conduct an entire session (*i.e.*, consisting in as many iterations as he wants). Our goal is to determine whether the system allows retrieval of interesting images, that is, thematically and visually relevant images. As a retrieval session may return numerous images, and as each image is ranked according to its expected visual and textual relevance, we say that a retrieval

session is "successful" whenever a certain amount of best ranked images are thematically and visually relevant.

We assume that images are presented to the user by groups of n images. We consider only the k first groups for evaluation. Each of these $k \times n$ images is evaluated by the user from a visual and textual point of view. Therefore, a symbolic grade is assigned to each image: A (very relevant), B (relevant), C (rather relevant), D (irrelevant), and E (totally irrelevant), according to its visual and textual user-relevance. Our decision rules are as follows: an image visually and thematically relevant gets grade A; an image visually or thematically relevant only gets grade C; finally, an image that is not thematically nor visually relevant gets grade E (of course, other decision rules may be adopted, depending on a particular search context).

We define a quality measure, noted Q, to evaluate the retrieval session performance, based on relevance judgements of the user (as described above), and on ranking:

$$Q = N \sum_{b=1}^{k} \left((k - b + 1) \sum_{p=1}^{n} v_{b,p} \right) \tag{1}$$

With:

b group number,

p image position within a group,

$v_{b,p}$ relevance value associated with image at position p in group b,

N normalisation factor.

Query Number	Query Formulation	Total nbr of relevant images
1	People at the airport	6
2	Military aircrafts	116
3	Sea birds	33
4	Boats	55
5	Soldiers in the desert	11
6	A cliff and a river	13
7	Cities views	70
8	Mountain sports	21
9	Trains or locomotives	43
10	Old cars	41
11	Fruits sales at the market	11

Table 1 Collection test.

The numerical value $v_{b,p}$ is derived from symbolic grades. In this experiment, we chose the values reported on table 2. Constant N may be set to 1, so that Q is a "general quality measure", or to $1/nk$, to obtain an "average quality measure".

A collection test of 11 queries has been defined, in order to evaluate system performance on the corpus. Table 1 lists these queries, along with a manual evaluation of the amount of potentially relevant images in the corpus.

Given the collection test described above, retrieval sessions have been conducted in different indexing conditions, namely 20%, 40% and *ad hoc*

Symbol	Value
A	+5
B	+3
C	0
D	-1
E	-2

Table 2 Numerical values for symbolic grades.

indexing. Only the 10 first images are taken into account (we consider 2 groups of 5 images, that is, $k = 2$ and $n = 5$ in equation 1), yielding a global performance measure Q. As the user may tune visual against textual weights to optimize the results, we also indicate the visual weight (in a range from 0 to 100). Tables 3 and 4 show the results corresponding to our 11 queries under different indexing conditions. The information provided in these tables corresponds to whole retrieval sessions, that is, after feedback, if any. It is organised as follows: for each query (col. 1), and under various indexing conditions — that is, 20%, 40% or ad hoc indexing — (col. 2), relevance judgements are reported for the 10 best ranked images (col. 3 to 12): "T" stands for "thematically relevant", "V" for "Visually relevant". A session quality score Q is computed according to equation 1, with $k = 2$, $n = 5$, $N = 1$ (col. 13). Finally, the visual weight used to produce the reported results is given (col. 14).

5.3 Discussion

In our approach, the *tunnel* mechanism has been devised to allow collaboration of textual and visual retrieval techniques. Experiments show that, for each query, an interesting set of images has been retrieved, by means of textual and/or visual similarity measures (see tables 3 and 4 for a complete report of our results, and tables 5 and 6 for a visual representation of our retrieval results for some queries).

Figure 4 reveals the contribution of textual, visual, and tunnel techniques. In this histogram, the average number of images retrieved using each technique is plotted. The lower part of the bar represent the number of images that are actually considered as thematically and visually relevant by user. We observe that the *tunnel* mechanism is fairly reliable: 62.5% of tunnel-retrieved images turned out to be relevant, whereas only 30% of images retrieved from visual techniques only proved to be relevant.

The hypothesis associated to the *tunnel* mechanism turned out to be verified: we observe that most of the images retrieved by means of *tunnel* have been judged as relevant by the user. In this way, the *tunnel* can be used to *confirm* image relevance. As a heuristic, we can say that an image retrieved by means of visual techniques *and* by means of *tunnel* is more likely to be relevant than another.

5.3.1 Query reformulation performance. We shall focus here on visual reformulation, that permitted retrieval results improvement. Global colour-

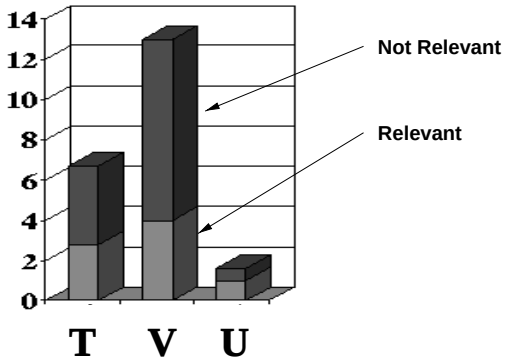


Fig. 4 Performance of (T)extual, (V)isual, and t(U)nnel retrieval techniques.

based reformulation allows the system to focus on a certain kind of images, namely those which are closer to the user's desires. As a complement, layout-based reformulation allows the system to refine image ranking (see results on table 6(a) and 6(b)).

Visual retrieval favours recall against precision. As a consequence, "noise" is the main drawback of this approach. We think, however, that this is necessary, as the corpus is not totally indexed. Actually, relevant images may not be indexed, and visual retrieval — combined with the *tunnel* mechanism — is the only chance we have to retrieve those images. Figure 5 gives an example of our results: whereas textual descriptions allows the retrieval of two interesting images only (step 1), the *tunnel* mechanism (step 2) acts as a relay and allows retrieval of 6 more images, that are not actually indexed. The virtual image and the relevance feedback process are responsible for improving image ranking, and therefore to allow those non-indexed images to be presented.

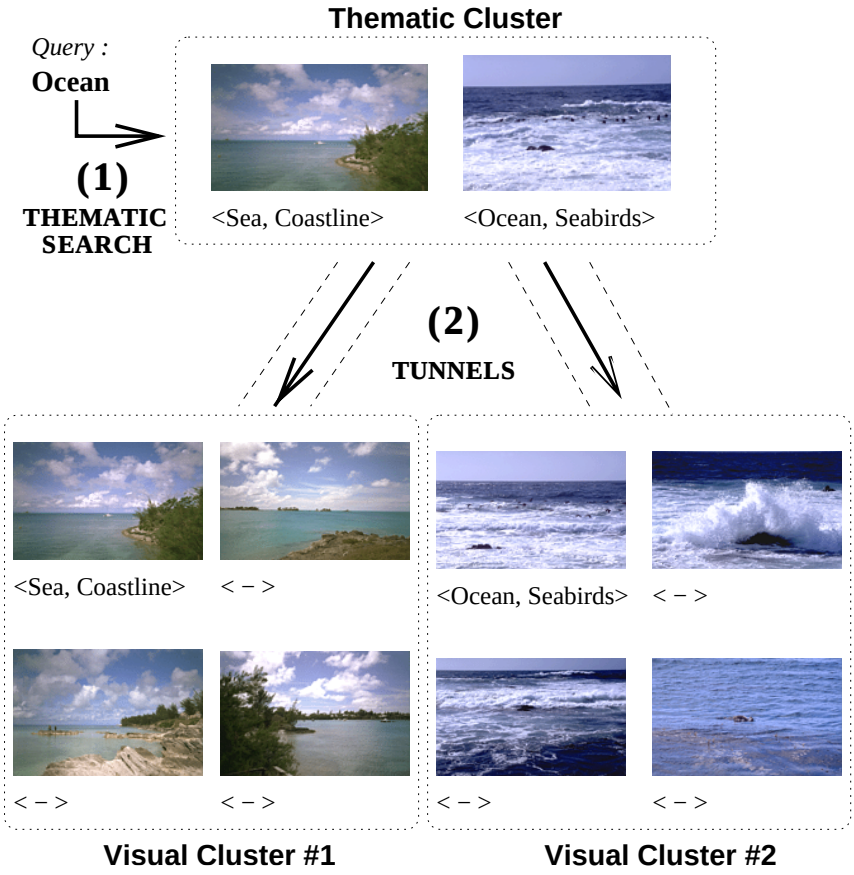


Fig. 5 An example of textual and visual retrieval results.

5.3.2 Pure visual similarity experiment. To evaluate the contribution of our textual and visual approach, we conducted an additional experiment to show in what extent pure visual retrieval could bring back interesting images on its own.

For each query, we chose to select among all relevant images an example that could be used to retrieved similar images in a "query by example" strategy, based only on visual similarity (*i.e.* the use of colour histogram intersection).

Figure 6 compares this pure visual retrieval to our textual and visual retrieval approach. For each query and each method, our quality measure is computed: vertical bars report pure visual retrieval score, where as the combined textual and visual score is represented by a cross (for readability purposes, crosses have been joined by a solid line). In two cases, visual retrieval achieved as good results as our combined method. In fact, very similar images existed in the database, and the colour histogram feature could easily retrieve them. In all other cases, the combined method yielded better results as:

- Many images can be relevant, although very different on a visual point of view. Query by visual example focuses on one possible representation.
- Global color histogram is a rough visual feature. We are dealing with complex images, that would require more accurate features.

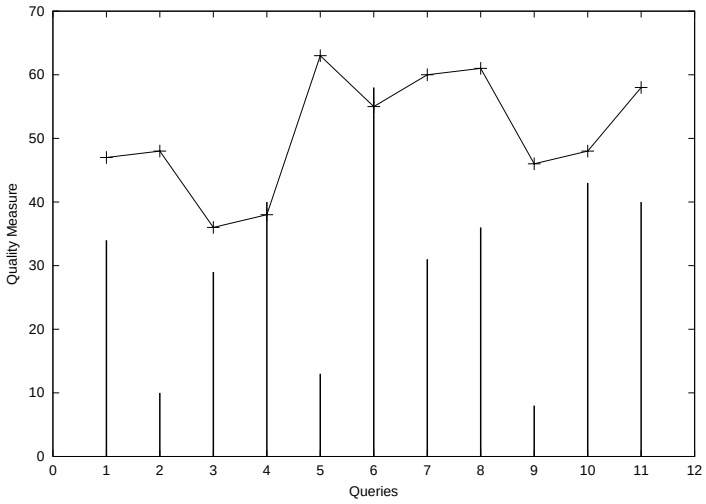


Fig. 6 Pure visual retrieval compared to textual and visual retrieval.

As a consequence, texture and shape features ought to be incorporated in an efficient visual similarity measure. This yields, however, to additional problems. Texture feature is only relevant for a small, homogeneous image clip: if various textures are present in this clip, then the texture feature will capture a "blurred" vision of this sub-image. When it comes to shape description, the difficult problem of shape identification arises. Before trying to characterize some shape, we need to determine what should be considered as a shape, that is, what is important to the user's eyes. This can vary, however, from a user to another, and it depends also on user's information need.

As we mentioned earlier, texture and shape features require some fine image segmentation to be efficient. We think however that this process can hardly be carried out automatically, as it depends on user's feeling, in a given retrieval situation. In our opinion, a possible solution is to allow him to draw onto an interesting image a region that he considers as relevant, and to further analyze this particular clip with more powerful visual features. This is to be implemented in a more sophisticated relevance feedback mechanism.

5.3.3 The impact of initial indexing. Initial textual indexing is of particular importance. We obtained the following average scores according to the three indexing conditions we considered: 35 at random 20%, 42 at random 40%, and 51 at *ad hoc* indexing. Figure 7 compares our results (detailed information is reported on tables 3 and 4). *Ad hoc* indexing achieved often the best results, even if far less images are actually indexed in this case.

Moreover, indexing more images does not systematically lead to better results.

It is clear that retrieval techniques based on visual similarity evaluation only are not sufficient, since visual indices are often not powerful enough to characterize accurately *all* kind of images. In this case, part of the corpus may be "invisible" to the system. Similar situation occurs when images are not indexed, and therefore not accessible from a textual point of view.

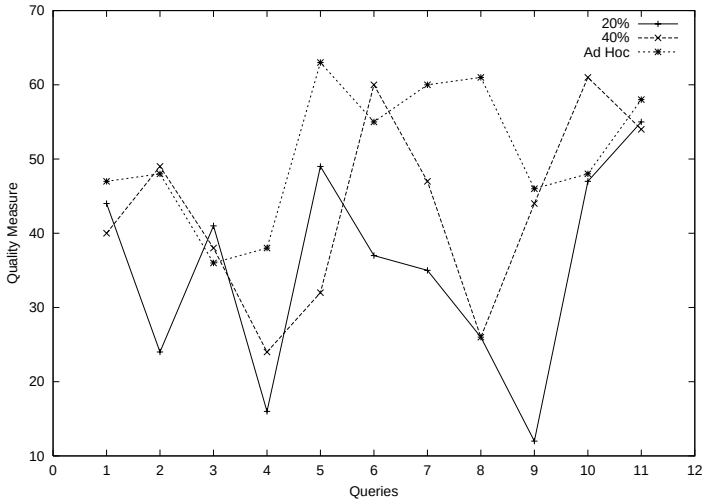


Fig. 7 Results under three different indexing conditions.

These remarks highlight the need for a textual and visual approach, that tries to avoid both of these situations. Indeed, the relevance feedback mechanism, based on both textual and visual aspects allows the user to choose what type of feedback (textual and/or visual) is to be applied at a particular step of his retrieval session. It is then possible to first focus on some very precise theme. During this stage, relevance feedback suggests more indexing

terms that could help retrieving new interesting images. Thereafter, the user can choose between different visually homogeneous classes of images, taking into account some visual expectations (this two-step process is illustrated in figure 5).

6 Conclusion and future work

We have presented an integrated method that allows textual and visual image retrieval in a partially indexed corpus. Our strategy relies on a preliminary corpus organisation into two hierarchical structures by a clustering process. An adapted retrieval process was designed that takes advantage of the user-system interactions: precise feedback data are gathered and processed within a *virtual image*. We also defined an original evaluation method suitable to the proposed approach.

This approach for image retrieval is well suited to large corpora in which a little proportion is textually indexed: visual index is computed off-line, fast image filtering is allowed by hierarchical clustering, and new images may be easily added to the existing hierarchies.

First experimentation results with the PICAP prototype confirm the relevance of our approach. We believe that PICAP could be improved by the integration of powerful visual characteristics which will make possible to refine not only the user query but also the similarity measure. This could be achieved by selecting the best characteristics according to the user and his preferences (see for example [2,18]), or by setting up a more sophisticated

relevance feedback mechanism. Moreover, interesting semantic relationships can be found in Wordnet, that could be used to refine our thematic relevance feedback. New relationships may be added to Wordnet, as suggested in [3], and should be considered.

Finally, we think that off-line session analysis is likely to produce interesting indexing hypothesises. Indeed, non-indexed images may have been retrieved during a session and user feedback is precise enough to allow us to determine if an image is relevant to a given query. Assuming some good properties of the query (such as compactness, i.e. few items are specified in the query), a temporary label can be attached to the image. Repeated successful retrievals of this image will increase the hypothesis confidence until it becomes a regular indexing item.

Furthermore, as initial visual clustering allows the identification of homogeneous images, we think that the *ad hoc* indexing process can be facilitated: each cluster is examined, and at least one image of a representative theme is manually indexed. Our process ensures that if this cluster is retrieved based on some textual query, all images in this cluster get a chance to be retrieved by means of the *tunnel* mechanism.

References

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Qry	Index	Image Ranks										Qual.	Vis.
Nbr	Cond.	1	2	3	4	5	6	7	8	9	10	Sc.	Wght
1	20%	TV	TV	TV	V	TV			TV	TV		44	70
	40%	TV	TV	TV	TV	TV						40	100
	5%	TV	TV	TV	TV	TV		TV				47	70
2	20%	TV	TV	T	TV	T	T			T		24	60
	40%	TV	TV	TV	TV	TV	TV	T				49	70
	5%	TV	TV	TV	TV	TV	T	T	T		T	48	90
3	20%	TV	TV	TV	TV	V	TV	V	V			41	70
	40%	TV	TV	TV	TV	V	V	V		T	T	38	30
	5%	TV	TV	T	TV	TV	T	T	T			36	40
4	20%	TV	V	T	V	TV	V		V	TV		16	60
	40%	TV	T	T	TV	TV	T				T	24	50
	5%	TV	TV	TV	TV	T		T	T	T	T	38	80
5	20%	TV	TV	TV	TV	TV	V		TV			49	100
	40%	TV	TV	TV	TV					T		32	50
	5%	TV	TV	TV	TV	TV	TV		TV	T	TV	63	70
6	20%	TV	TV	TV	TV				TV			37	100
	40%	TV	TV	TV	TV	TV	TV	V	V	V	TV	60	100
	5%	TV	TV	TV	TV	TV	TV	T	V	V	V	55	60

Table 3 Query results (5% indexing is *ad hoc* indexing).

2. G. Ciocca and R. Schettini. A relevance feedback mechanism for content-based image retrieval. *Information Processing and Management*, 35:605–632, 1999.

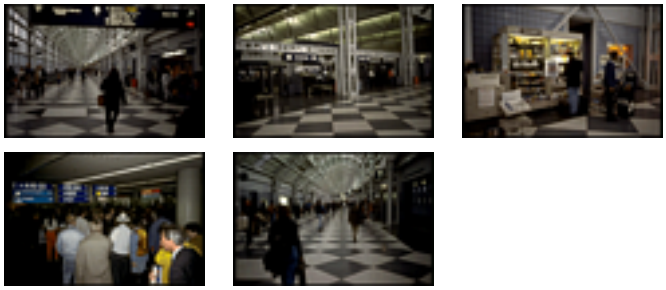
Qry	Index	Image Ranks										Qual.	Vis.
Nbr	Cond.	1	2	3	4	5	6	7	8	9	10	Sc.	Wght
7	20%	TV	TV	V	TV	TV	V	T				34	90
	40%	TV	TV	TV	TV	TV					TV	47	100
	5%	TV	TV	TV	TV	TV	TV	T	V	TV	V	60	60
8	20%	TV	TV	TV	TV							26	60
	40%	TV	TV	TV		TV						26	70
	5%	TV	TV	TV	TV	TV	TV	TV			TV	61	100
9	20%	TV	TV	TV								12	60
	40%	TV	TV	TV	TV	TV		T			T	44	70
	5%	TV	TV	TV	TV	T	TV		TV	T		46	20
10	20%	TV	TV		TV	TV	TV			TV	TV	47	95
	40%	TV	TV	TV	TV	TV	TV	TV	TV			61	60
	5%	TV	TV	TV	TV	TV	T	T	T		T	48	70
11	20%	TV	TV	TV	TV	TV	TV	V	T	V	V	55	90
	40%	TV	TV	TV	TV	TV		TV	TV			54	50
	5%	TV	TV	TV	TV	TV	TV	TV		T	V	58	100

Table 4 Query results (5% indexing is *ad hoc* indexing).

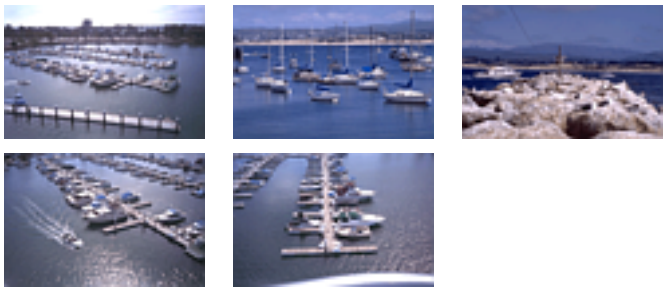
3. F. Dehne, A. Steuten, and R. van de Riet. WordNet++: A lexicon for the Color-X-method. *Data & Knowledge Engineering*, 38:3–29, 2001.

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(a) Query #1 (*ad hoc* indexing)



(b) Query #4 (40% indexing)

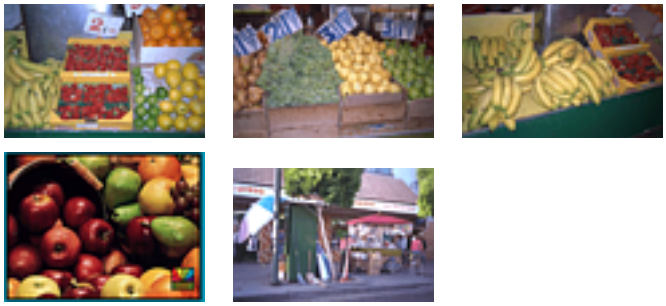


(c) Query #9 (40% indexing)

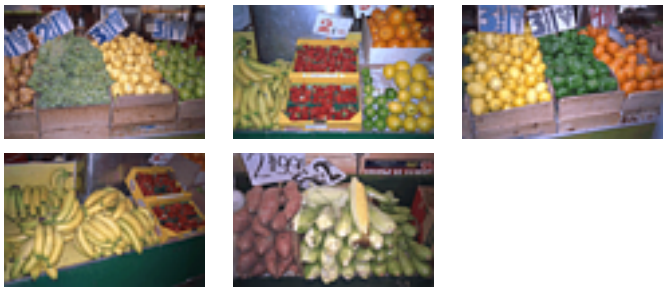
Table 5 Results for queries 1, 4 and 9.

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(a) Query # 11 (20% indexing) without layout feedback



(b) Query # 11 (20% indexing) after layout feedback

Table 6 Results for queries 9 and 11.

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